

STAT 341: Tutorial Test 2 - Sample 1

First name

Last name

Signature

ID Number

Email

Time allowed: 50 minutes

Special instructions

- All numeric answers should be written either as a number to three decimal places or as a fraction reduced to its lowest terms.
- If working is not specifically requested, a correct answer will receive full marks. However, an incorrect answer could still receive credit if working is provided. An incorrect answer with no working will receive zero marks.

Please read: this practice test is designed to give you a clear sense of the sort of things that can come up in exams. Please note that this is, of course, only a subset of the material that could be covered. I would **strongly** encourage you to attempt this test under exam conditions - do not be tempted to look up something in your notes until afterwards! This is one reason why it is common for students to find real exams much harder than practice ones, even though the material is of an equivalent difficulty!

1. Consider estimating the mean of the population $\mathcal{P} = \{1, 2, 3, 4\}$ based on a random sample of size 2 (without replacement). The sampling protocol (probability of all possible samples) is summarized in the table below.

S	$\{1, 2\}$	$\{1, 3\}$	$\{1, 4\}$	$\{2, 3\}$	$\{2, 4\}$	$\{3, 4\}$
$P(S)$	0.15	0.25	0.1	0	0.2	0.3

- (a) [2 marks] What is sampling bias for the sample mean, $\bar{Y}$.

$$\bar{y}_{\{1,2\}} = 1.5, \quad \bar{y}_{\{1,3\}} = 2, \quad \bar{y}_{\{1,4\}} = 2.5, \quad \bar{y}_{\{2,3\}} = 2.5, \quad \bar{y}_{\{2,4\}} = 3, \quad \bar{y}_{\{3,4\}} = 3.5$$

$$E[\bar{Y}] = \sum_{\substack{u \text{ is a} \\ \text{sample}}} \bar{y}_u P(\bar{Y} = \bar{y}_u) = 1.5(0.15) + 2(0.25) + 2.5(0.1) + 2.5(0) + 3(0.2) + 3.5(0.3) \\ = 2.625$$

$$\text{Population mean} = \mu = (1+2+3+4)/4 = 2.5$$

$$\therefore \text{Sampling bias} = 2.625 - 2.5 = 0.125 //$$

- (b) [2 marks] Calculate the Horvitz-Thompson estimate of the population mean based on the sample $S = \{2, 4\}$.

The inclusion probabilities of 2 and 4 are

$$\pi_2 = 0.15 + 0 + 0.2 = 0.35$$

$$\pi_4 = 0.1 + 0.2 + 0.3 = 0.6$$

$$\Rightarrow \bar{y}_{\{2,4\}, HT} = \frac{1}{4} \left(\frac{2}{0.35} + \frac{4}{0.6} \right) = 3.095 //$$

2. [2 marks] Suppose we are interested in the average length of sharks in great white shark encounters with humans in US waters but have access only to those encounters which were in Australian waters.

- The **target population** is

Great white sharks in US waters //

- The **study population** is

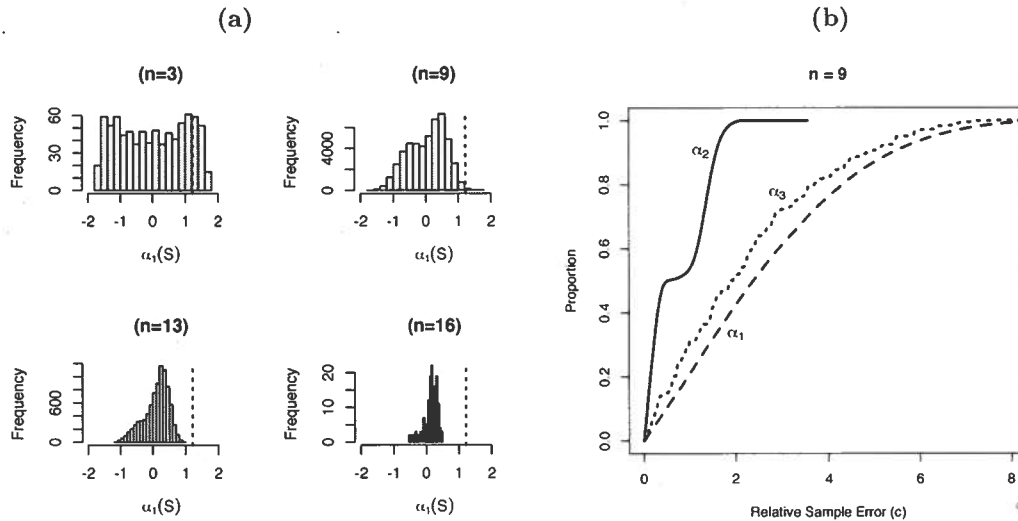
Great white sharks in Australian waters //

3. [2 marks] Every measuring system has sources of potential error. Name two sources of potential measurement error.

Any two of:

1. the measuring device
2. the person reading or recording the measurement
3. the method followed to take the measurement. //

4. Consider a population $\mathcal{P} = \{u_1, \dots, u_{18}\}$ from which a simple random sample of size n without replacement is taken. Interest lies in studying the skewness in this population. Based on all possible samples of sizes $n = 3$, $n = 9$, $n = 13$, and $n = 16$, panel (a) plots histograms of a skewness attribute $\alpha_1(S)$ for different sample sizes. Superimposed in black dashed lines on the histograms is the population value $\alpha_1(\mathcal{P})$. Beside α_1 , two other skewness attributes α_2 and α_3 exist in statistics literature. Panel (b) plots the consistency curves for these three attributes. In particular, given a value c , each curve represents the proportion of samples with relative sample error below c .



(a) [2 marks] What do the plots in panel (a) tell you about the estimator α_1 ?

- The distribution of the estimates $\alpha_1(S)$ becomes more concentrated as the sample size increases.
- $\alpha_1(S)$ is a biased estimator of $\alpha_1(\mathcal{P})$, as we can see a negative sampling bias. //

(b) [2 marks] What do the plots in panel (b) tell you about the estimators α_1 , α_2 and α_3 ?

- For any given c , the proportion of samples with relative sample error below c is the highest for α_2 , then α_3 , followed by α_1 .
- α_2 is the best estimator among the three. //

5. (a) [4 marks] Describe in point form a test of significance using a discrepancy measure $D(P_1, P_2)$

1. Generate a hypothesis: P_1 & P_2 are drawn from the same population.
2. Construct a discrepancy measure $D(P_1, P_2)$, where large values indicate evidence against the hypothesis.
3. Compute d , the observed discrepancy measure.
4. Compute the probability of $D \geq d$ when the hypothesis is true. //

(b) [1 mark] Why is $D(P_1, P_2) = |a(P_1) - a(P_2)|$ a discrepancy measure?

Assume that $a(\cdot)$ is such that, the more different P_1 and P_2 are, the more different $a(P_1)$ and $a(P_2)$ are.

Then, for the hypothesis that P_1 and P_2 are from the same population, large values of $D(P_1, P_2)$ indicate evidence against the hypothesis. //

6. [6 marks (1 mark each)] In the statements below, complete the sentence or determine if the statement is true or false.

(a) There exist a total of^{6³} possible samples, each of size 3, when taking a simple random sample with replacement from the population $\mathcal{P} = \{u_1, u_2, u_3, u_4, u_5, u_6\}$.

(b) The Horvitz-Thompson estimator of the population mean is a scale equivariant estimator.

- i. True
- ii. False

(c) The difference between the attribute of the study population and that of the sample is called the study error.

- i. True
- ii. False

(d) If the sampling protocol is simple random sampling without replacement, then the sample average is also the Horvitz-Thompson estimator of population mean.

- i. True
- ii. False

(e) The relative error of an estimator is a number between 0 and 1.

- i. True
- ii. False

(f) Scale invariance is a desirable property for measures of spread in a population.

- i. True
- ii. False